D 138 | Test a segment, not a flattering story

PRIVATE FICTIONAL adult packaged-snack research design. No nutrition advice, actual outreach, interviews, dataset download, participants, quotes, consent or market validation.

Pilot comparison	A: 100 qualified offers, 12 purchases, price 200, variable cost 120, acquisition 600. B: 200 offers, 18 purchases, unit contribution 50, acquisition 300. Conversion A 12%, B 9%; net contribution A 360, B 600. A net per offer 3.6 exceeds B 3.0, despite smaller cohort total. No randomized causal proof.
Costs and denominators	Contribution after attributable acquisition is not accounting profit, revenue or full customer lifetime value. No fixed costs, refunds, retention, capacity or nutrition outcomes supplied. Cost per purchase undefined at zero purchases. Conversion undefined at zero offers, not zero.
Relative search interest	Fictional A 200 matching / 10000 total searches = 2%; B 1000 / 100000 = 1%. Shared maximum 2% => indices 100 and 50. B has five times the matching searches, not five times relative interest. No actual Trends data downloaded.
Do not fabricate scale	Google Trends uses sampled, normalized search requests and statistical noise; not a scientific poll, population share, absolute demand or sales. Low-volume 0 and blank maps do not prove no searches or demand. New maxima can rescale every index without changing raw search shares.
Actionable hypotheses	Need/occasion is a hypothesis, not an automatically superior label. Demographic/geographic cuts can work if responses differ and delivery is feasible. ADAMS asks accessible, differentiable, actionable, measurable, substantial; toy flags are not empirical proof.
Interview plan only	Write three falsifiable segment hypotheses and two blank interview protocols: recent episode, current alternative, tradeoff, constraint, disconfirmation and next step. Intent is qualified evidence, not no evidence or a purchase. No outreach or invented quotes/commitments. Consent and anonymization required if later authorized.

Inputs | Synthetic pilot and search controls

Yellow values are fictional. Offers, purchases and search counts are whole/nonnegative; purchases <= offers; matching <= total. Monetary values nonnegative. No actual purchase or ad spend.

A qualified offers	100.00
A purchases	12.00
A price per purchase	200.00
A variable cost per purchase	120.00
A attributable acquisition spend	600.00

B qualified offers	200.00
B purchases	18.00
B unit contribution before acquisition	50.00
B attributable acquisition spend	300.00

A matching searches	200.00
A total searches	10000.00
B matching searches	1000.00
B total searches	100000.00

Inputs | Synthetic pilot and search controls

Yellow values are fictional. Offers, purchases and search counts are whole/nonnegative; purchases <= offers; matching <= total. Monetary values nonnegative. No actual purchase or ad spend.

Toy: accessible evidence? 0/1	1.00
Toy: differentiable evidence? 0/1	1.00
Toy: actionable evidence? 0/1	0.00
Toy: measurable evidence? 0/1	1.00
Toy: substantial evidence? 0/1	1.00

Pilot A controls valid?	1.000000
Pilot B controls valid?	1.000000
Search-count controls valid?	1.000000
Toy ADAMS controls valid?	1.000000

Zero valid counts allowed but undefined ratios stay #N/A. Negative unit contribution for A is permitted when variable cost exceeds price; underlying price/cost remain nonnegative. Shared search maximum is derived from current shares; all-zero shares have no defined index.

Pilot A | Denominators and contribution

Currency units. Contribution after acquisition is not revenue, accounting profit, full CAC or LTV.

Conversion fraction	0.120000
Unit contribution before acquisition	80.000000
Revenue	2400.000000
Contribution before acquisition	960.000000
Contribution after acquisition	360.000000
Net contribution per qualified offer	3.600000
Attributable acquisition spend per purchase	50.000000

Baseline 0.12, 80, 2400, 960, 360, 3.6, 50. Zero purchases leaves acquisition per purchase undefined but net contribution = negative acquisition spend.
Variable cost above price makes unit contribution negative; no arbitrary clipping to zero.

Pilot B | Cohort total is not per-offer superiority

B supplies unit contribution directly, not price or variable cost. Revenue cannot be reconstructed.

Conversion fraction	0.090000
Contribution before acquisition	900.000000
Contribution after acquisition	600.000000
Net contribution per qualified offer	3.000000
Attributable acquisition spend per purchase	16.666667

Baseline 0.09, 900, 600, 3, 16.666667. A has higher conversion and net per offer; B higher cohort total. Unequal unrandomized cohorts do not identify causality or a market-wide forecast. No scale, retention or full-cost assumption supplied.

Search index | Normalize shares, not matching counts

Synthetic arithmetic only. Compatible term/topic, time, geography, category and property must be documented for any later authorized comparison.

A matching share	0.020000
B matching share	0.010000
Shared maximum matching share	0.020000
A index on derived shared maximum	100.000000
B index on derived shared maximum	50.000000
B/A raw matching-count ratio	5.000000

Baseline 0.02, 0.01, 0.02, 100, 50, 5. Doubling both A counts keeps share/index unchanged; increasing B share can rescale A index. Separate charts each peaking 100 do not establish equal volumes. Actual provider noise/sampling/rounding not simulated; this is not the full Trends algorithm.

ADAMS | Actionability gap is not solved by a label

Toy evidence switches, not independent empirical validation, authority to launch, a chosen customer or a market-size estimate.

Accessible evidence?	1.000000
Differentiable evidence?	1.000000
Actionable evidence?	0.000000
Measurable evidence?	1.000000
Substantial evidence?	1.000000
All listed toy evidence switches pass?	0.000000

Baseline 1/1/0/1/1=>0. All 1 only passes fictional checklist, not product-market fit, nutrition safety, legal compliance or real evidence. No automatic biggest/smallest/highest-index choice. State an uncertainty, next evidence and stop condition.

Research plan | Falsifiable needs and neutral questions

Model prompts only, no participants, quotes, purchase commitments or completed interviews.

Three hypotheses	For each: adult occasion/need, current alternative, constraint, reach route, plausible economic value and observation that would overturn it. A narrow label does not prove unmet need or segment homogeneity.
Recent behavior	Neutral opening: Tell me about the last time you bought a snack for that occasion. Follow actual sequence, alternative, cost/time tradeoff and constraint. Do not pitch an affordable healthier snack as a leading question.
Evidence classification	Separate recalled episode, observed behavior, stated intention and concrete next step. Prior spending is a useful reference, not proven willingness to pay for a new offer. Conditional intent is weaker evidence, not automatically worthless.
Two protocols, not interviews	Use two blank protocol rows. If later authorized: obtain consent, avoid unnecessary identifiers, anonymize notes, respect withdrawal and keep direct quotes exact. Two convenience interviews cannot estimate population prevalence.
Future Trends comparison	Record term versus topic, quoted phrase when intended, language scope, geography/time/category/property and common comparison normalization. Terms do not automatically include synonyms/spelling/plurals; topics group a concept across languages.
Beachhead decision	Compare reach, response differences, ability to deliver, worthwhile scale and economics. Preserve conflicting evidence. Choose a provisional test and stop condition only; no live launch, ad purchase or outreach authorized.

Actual work | Blank hypotheses and interview protocols

PRIVATE planning only. No actual interview, consent, quote, customer, participant or Trends dataset has been obtained.

Hypothesis	Alternative	Constraint	Reach / action	Disconfirming test	Economics gap	Stop condition

Protocol 1: recent behavior / alternative / constraint
Protocol 2: recent behavior / alternative / constraint
Future comparison filters / source / uncertainty
Provisional beachhead / evidence / stop

No invented interviews or quotes. Consent status remains unfilled, not assumed granted. Change one family, recompute, explain and restore. Original exercise/quiz preserved. All costs/counts fictional; no real spend or outreach.

Evidence | Written coverage, not field validation

Six URLs from four organizations: educational chapter, official provider help, author teaching resources and a limited practitioner excerpt, not independent efficacy tests.

OpenStax 5.4	https://openstax.org/books/principles-marketing/pages/5-4-essential-factors-in-effective-market-segmentation | Complete returned written chapter/knowledge checks read. ADAMS criteria; ideal homogeneity not an empirical guarantee for a proposed segment.
Google FAQ	https://support.google.com/trends/answer/4365533 | Complete returned written FAQ read: sample, denominator normalization, 0-100, noise, low-volume 0, not polling. Official help, not independent evidence of actual snack demand.
Google compare	https://support.google.com/trends/answer/4359550?hl=en | Complete returned written comparison help read. Terms, quoted phrases, topics, filters; current help distinguishes new versus classic interface limits. No UI flow or actual dataset verified.
Google regional	https://support.google.com/trends/answer/4355212 | Complete returned written regional help read. Relative share, not volume; blank region is not evidence of no interest. No map/data inspected.
Mom Test teachers	https://www.momtestbook.com/teachers | Complete returned teacher page read: bad/good questions, biased data, commitments and mock-conversation exercises. Not full book or classroom/field efficacy verification. Purchase terms not used.
HBR 2006 excerpt	https://hbr.org/2006/02/rediscovering-market-segmentation | Returned opening/summary excerpt only, not full article. Behavior/change/unmet needs versus advertising profiles; not a ban on useful demographics. No AV, native cross-viewer, independent learner/account/field/nutrition/legal or overall 10 claim.

Failure log | Authored checks, not a market verdict

Changed/invalid controls, broken references and clean restore. Actual research and learner transfer remain unverified.

1	PASS: 28 baseline formulas independently recomputed.
2	PASS: 24 changed sets plus clean restore.
3	PASS: 16 invalid controls/448 isolation checks.
4	PASS: 6 broken references/6 explicit error checks.
5	PASS: zero offers/purchases keep ratios undefined.
6	PASS: all-zero search shares give undefined indices.
7	PASS: negative A unit contribution retained, not clipped.
8	PASS: actual hypotheses and protocols blank.